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Binary Darboux transformations for a skew-symmetric matrix integrable nonlinear Schrödinger equation

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ABSTRACT

The aim of this article is to construct a class of binary Darboux transformations for a skew-symmetric matrix integrable nonlinear Schrödinger equation. The associated AKNS-type Lax pairs, corresponding to the orthogonal Lie algebra $so(2m)$, provide the framework necessary for systematically building these binary Darboux transformations. Direct, single-step applications of the transformations produce explicit soliton solutions. In particular, the M -matrix, central to the formulation of the binary Darboux transformations, is constructed within an extended framework that accommodates cases in which eigenvalues coincide with adjoint eigenvalues, thus incorporating the generalized Darboux transformations.

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1. Introduction

Integrable models provide an essential framework for the study of important coherent structures, such as solitons, breathers, lump waves, and rogue waves. These systems are typically associated with Lax pairs, which consist of spatial and temporal matrix spectral problems [1]. Building on such Lax pairs, numerous analytical methods have been developed for solving integrable models, including Darboux transformations, the inverse scattering transform, and the Riemann–Hilbert approach [2–4].

Let $u = u(x, t)$ denote the potential and λ the spectral parameter. For an integrable model, suppose that two linear spectral problems define its Lax pair:

$$\phi_x = U\phi = U(u, \lambda)\phi, \phi_t = V\phi = V(u, \lambda)\phi, \quad (1)$$

where U and V are square matrices. The Lax pair determines an integrable model via the compatibility condition $\phi_{xt} = \phi_{tx}$, which is equivalent to the zero-curvature equation [5,6]:

$$U_t - V_x + [U, V] = 0. \quad (2)$$

From the adjoint point of view, the corresponding adjoint matrix spectral problems are given by

$$\psi_x = -\psi U, \quad \psi_t = -\psi V, \quad (3)$$

whose compatibility condition $\psi_{xt} = \psi_{tx}$ leads to the same zero-curvature equation, and hence the same integrable model.

This paper considers an integrable nonlinear Schrödinger (NLS) equation with two skew-symmetric matrix potentials p and q :

$$p_t = \frac{1}{2}(p_{xx} - 2pqp), \quad q_t = -\frac{1}{2}(q_{xx} - 2qpq), \quad p^T = -p, \quad q^T = -q. \quad (4)$$

This system is generated from the following Lax pair associated with the orthogonal Lie algebra $\mathfrak{so}(2m)$:

$$U = \begin{bmatrix} \lambda I_m & p \\ q & -\lambda I_m \end{bmatrix}, \quad V = \lambda^2 \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix} + \lambda \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} pq & -p_x \\ q_x & -qp \end{bmatrix}, \quad (5)$$

where p and q are $m \times m$ skew-symmetric matrices, and I_m denotes the $m \times m$ identity matrix.

We focus on constructing binary Darboux transformations (DTs) for this skew-symmetric matrix integrable NLS equation. A binary DT is implemented via a gauge transformation (see, e.g. [7]):

$$\phi' = S^+ \phi, \quad \psi' = \psi S^-, \quad S^+ = S^+(u, \lambda), \quad S^- = S^-(u, \lambda), \quad (6)$$

where S^+ and S^- are mutual inverses and can be derived from a so-called M -matrix formulation (see, e.g. [7] for a general framework). The transformation conditions are given by

$$\begin{cases} S_x^+(S^+)^{-1} + S^+ U (S^+)^{-1} = U(u', \lambda), \\ S_t^+(S^+)^{-1} + S^+ V (S^+)^{-1} = V(u', \lambda), \end{cases} \quad (7)$$

where the transformed potential u' is defined by

$$u' = f(u). \quad (8)$$

Accordingly, ϕ' and ψ' form a new pair of eigenfunction and adjoint eigenfunction satisfying

$$\phi'_x = U' \phi', \quad \phi'_t = V' \phi', \quad (9)$$

and

$$\psi'_x = -\psi' U', \quad \psi'_t = -\psi' V', \quad (10)$$

where

$$U' = U(u', \lambda), \quad V' = V(u', \lambda). \quad (11)$$

Thus, the transformed Lax pair (U', V') satisfies a zero-curvature equation of the same structure as the original one, with u replaced by u' . Consequently, the transformed potential u' provides a new solution to the integrable model determined by (U, V) .

In modern soliton theory, binary DTs have been widely applied to a variety of integrable models, including the NLS equation and the modified Korteweg–de Vries equation (see, e.g. [8–10]), particularly for generating N -soliton solutions, which include breather, rogue, and lump wave solutions as special cases. In many situations involving Lax pairs with specific structures, it is necessary to impose orthogonality conditions between eigenfunctions and adjoint eigenfunctions (see, e.g. [8]), especially when eigenvalues coincide with adjoint eigenvalues.

The aim of this paper is to establish a class of binary DTs for the skew-symmetric matrix integrable NLS equation introduced above. The resulting transformations are constructed via an extended M -matrix generating procedure, which accommodates cases where eigenvalues coincide with adjoint eigenvalues. Single applications of the proposed transformations yield explicit soliton solutions. The paper concludes with a summary of the main results and final remarks.

2. A skew-symmetric matrix integrable NLS equation

Let m be an arbitrary natural number. We consider the following coupled system of integrable NLS equations:

$$\begin{cases} p_t = \frac{1}{2}p_{xx} - pqp, \\ q_t = -\frac{1}{2}q_{xx} + qpq, \end{cases} \quad (12)$$

where p and q are two $m \times m$ skew-symmetric matrices. This system is generated from the Lax pair of matrix spectral problems:

$$\phi_x = U\phi, \quad \phi_t = V\phi, \quad (13)$$

where U and V are associated with the orthogonal Lie algebra $\mathfrak{so}(2m)$ and are given by (5), namely,

$$U = U(u, \lambda) = \lambda\Lambda + P, \quad (14)$$

and

$$V = V(u, \lambda) = \lambda^2\Lambda + \lambda P - \frac{1}{2}\Lambda(P^2 - P_x), \quad (15)$$

where

$$\Lambda = \begin{bmatrix} I_m & 0 \\ 0 & -I_m \end{bmatrix}, \quad P = \begin{bmatrix} 0 & p \\ q & 0 \end{bmatrix}, \quad (16)$$

and $u = u(p, q)$. The skew-symmetric conditions

$$p^T = -p, \quad q^T = -q, \quad (17)$$

are equivalent to the reduction

$$\Sigma P \Sigma^{-1} = -P^T, \quad \Sigma = \begin{bmatrix} 0 & I_m \\ I_m & 0 \end{bmatrix}, \quad (18)$$

which ensures that P , and hence U and V , belong to the orthogonal Lie algebra $\mathfrak{so}(2m)$.

Associated with the above spatial spectral matrix U , there exists an infinite number of commuting flows:

$$u_t = X^{[k]} = X^{[k]}(u) = \begin{bmatrix} 2b^{[k+1]} \\ -2c^{[k+1]} \end{bmatrix}, \quad k \geq 0, \quad (19)$$

which are generated from the Lax operators $V^{[k]}$:

$$V^{[k]} = \sum_{l=0}^k \lambda^{k-l} W^{[l]}, \quad W^{[l]} = \begin{bmatrix} a^{[l]} & b^{[l]} \\ c^{[l]} & d^{[l]} \end{bmatrix}. \quad (20)$$

Here, the Laurent series

$$W = \sum_{l \geq 0} W^{[l]} \lambda^{-l}, \quad W^{[0]} = \Lambda, \quad (21)$$

is assumed to solve the stationary zero-curvature equation (see, e.g. [11]):

$$W_x = [U, W]. \quad (22)$$

The hierarchy is uniquely determined by imposing the normalization condition

$$W^{[l]}|_{u=0} = 0, \quad l \geq 1. \quad (23)$$

In particular, the member corresponding to $k=2$ coincides with the matrix integrable NLS model (12), since

$$V = \lambda^2 W^{[0]} + \lambda W^{[1]} + W^{[2]}. \quad (24)$$

The commutativity of these flows,

$$[[X^{[k]}, X^{[l]}]] := X^{[k]'}(u)[X^{[l]}] - X^{[l]'}(u)[X^{[k]}] = 0, \quad k, l \geq 0, \quad (25)$$

can be verified using the iso-spectral Lax operator algebra (see, e.g. [12,13]):

$$[[V^{[k]}, V^{[l]}]] := V^{[k]'}(u)[X^{[l]}] - V^{[l]'}(u)[X^{[k]}] + [V^{[k]}, V^{[l]}] = 0, \quad k, l \geq 0, \quad (26)$$

By imposing suitable reductions, the integrable NLS model (12) can be reduced to several typical coupled integrable models, thereby extending the class of multi-component NLS-type systems studied in the literature (see, e.g. [14–19]). In particular, under the reduction

$$p(t, x) = \frac{\sqrt{2}}{2} v \left(\frac{1}{2} it, ix \right), \quad q(t, x) = \pm \frac{\sqrt{2}}{2} v^\dagger \left(\frac{1}{2} it, ix \right), \quad (27)$$

where † denotes the Hermitian conjugate, the model (12) reduces to the matrix focusing or defocusing NLS equation:

$$iv_t + v_{xx} \pm vv^\dagger v = 0, \quad (28)$$

where v is a square matrix. Furthermore, when

$$v = \begin{bmatrix} 0 & w \\ -w & 0 \end{bmatrix},$$

with w being a scalar function, the matrix NLS equation further reduces to the standard scalar focusing and defocusing NLS equation:

$$iw_t + w_{xx} \pm |w|^2 w = 0, \quad (29)$$

which exhibits significant applications in areas such as nonlinear optics [20].

3. Binary Darboux transformations

We consider the skew-symmetric matrix integrable NLS equation (12), together with its associated Lax pair defined in (13), where the matrices U and V are given by (14) and (15), respectively.

The aim of this section is to construct a class of binary DTs based on this Lax pair formulation. To incorporate both eigenfunctions and adjoint eigenfunctions, we develop an extended M -matrix framework for generating DTs for the matrix integrable NLS equation (12).

In the first subsection, we establish a binary DT formulation for the general linear Lie algebra $\mathfrak{gl}(2m)$. In the second subsection, we extend this construction to the orthogonal Lie algebra $\mathfrak{so}(2m)$.

3.1. The unreduced case associated with $\mathfrak{gl}(2m)$

Let u be a given solution of the integrable NLS model (12), with two unreduced matrix potentials p and q . Suppose that N is an arbitrary natural number, and that we construct a class of binary DTs based on N pairs of eigenfunctions and adjoint eigenfunctions.

Assume that λ_i and $\hat{\lambda}_i$ ($1 \leq i \leq N$) are eigenvalues and adjoint eigenvalues, respectively, and that v_i and $\hat{v}_i$ are the corresponding eigenfunctions and adjoint eigenfunctions. Then, we have

$$v_{i,x} = U(u, \lambda_i)v_i, \quad v_{i,t} = V(u, \lambda_i)v_i, \quad 1 \leq i \leq N, \quad (30)$$

$$\hat{v}_{i,x} = -\hat{v}_i U(u, \hat{\lambda}_i), \quad \hat{v}_{i,t} = -\hat{v}_i V(u, \hat{\lambda}_i), \quad 1 \leq i \leq N. \quad (31)$$

We will use these eigenfunctions and adjoint eigenfunctions to establish a formulation for N -fold binary DTs. This framework enables the generation of new solutions to the matrix integrable NLS equation (12) from known ones.

To construct binary DTs, we consider

$$\begin{cases} \Theta_{ij} := \frac{U(u, \lambda_j) - U(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = \Lambda, \\ \Xi_{ij} := \frac{V(u, \lambda_j) - V(u, \hat{\lambda}_i)}{\lambda_j - \hat{\lambda}_i} = (\lambda_j + \hat{\lambda}_i)\Lambda + P. \end{cases} \quad (32)$$

A crucial step in constructing the N -fold binary DTs is the introduction of an $N \times N$ matrix $M = (m_{ij})$, based on these quantities. To encode the interactions between the eigenfunctions and adjoint eigenfunctions (see, e.g. [10]), we define

$$M = (m_{ij})_{N \times N}, \quad m_{ij} = \begin{cases} \frac{\hat{v}_i v_j}{\lambda_j - \hat{\lambda}_i}, & \text{if } \lambda_j \neq \hat{\lambda}_i, \\ m_{ij}^c(x, t), & \text{if } \lambda_j = \hat{\lambda}_i, \end{cases} \quad \text{where } 1 \leq i, j \leq N, \quad (33)$$

where v_i and $\hat{v}_i$ are the eigenfunctions and adjoint eigenfunctions introduced in (30) and (31). The entries $m_{ij}^c(x, t)$ correspond to the coinciding-eigenvalue case $\lambda_j = \hat{\lambda}_i$, and are defined by

$$m_{ij,x}^c = \hat{v}_i \Theta_{ij} v_j, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad (34)$$

and

$$m_{ij,t}^c = \hat{v}_i \Xi_{ij} v_j, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad (35)$$

for $1 \leq i, j \leq N$. This M -matrix plays a central role in constructing the higher-rank projector that defines the binary Darboux matrices. In particular, it encompasses generalized DTs in which eigenvalues may coincide with adjoint eigenvalues.

As soon as the $N \times N$ matrix M defined in (33) is invertible, we can formulate the N -fold binary DTs as follows. More precisely, the Darboux matrices are defined by

$$S^+ = S^+(\lambda) = I_{2m} - \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \hat{\lambda}_j}, \quad S^- = S^-(\lambda) = I_{2m} + \sum_{i,j=1}^N \frac{v_i (M^{-1})_{ij} \hat{v}_j}{\lambda - \lambda_i}, \quad (36)$$

together with their leading-order terms at infinity:

$$S_1^\pm = \lim_{\lambda \rightarrow \infty} [\lambda (S^\pm(\lambda) - I_{2m})]. \quad (37)$$

In the above formulation, v_i and $\hat{v}_j$ are the eigenfunctions and adjoint eigenfunctions defined in (30) and (31), respectively.

On the basis of the general binary DT theory, we need to verify the following two key conditions:

$$S_x^+ = U' S^+ - S^+ U, \quad (38)$$

and

$$S_t^+ = V' S^+ - S^+ V. \quad (39)$$

Here, the new Lax pair (U', V') is defined by

$$U' = \lambda \Lambda + P', \quad V' = \lambda^2 \Lambda + \lambda P' - \frac{1}{2} \Lambda (P'^2 - P'_x), \quad (40)$$

where P' is given by

$$P' = \begin{bmatrix} 0 & p' \\ q' & 0 \end{bmatrix} = P + S_1^+ \Lambda + \Lambda S_1^-. \quad (41)$$

Thus, the formulation of a class of binary DTs for the matrix integrable NLS equation (12) reduces to verifying the two conditions (38) and (39).

We point out that when the eigenvalues satisfy the following non-degeneracy conditions:

$$\begin{cases} \{\lambda_i \mid 1 \leq i \leq N\} \cap \{\hat{\lambda}_i \mid 1 \leq i \leq N\} = \emptyset; \\ \lambda_i \neq \lambda_j, \quad 1 \leq i \neq j \leq N; \\ \hat{\lambda}_i \neq \hat{\lambda}_j, \quad 1 \leq i \neq j \leq N; \end{cases} \quad (42)$$

then (36) reduces to Darboux matrices in the standard N -fold DT formulation, which has been extensively studied in the literature (see, e.g. [3,4,21–25]). If some eigenvalues coincide with adjoint eigenvalues, then the matrix M must be constructed with care, and the resulting formulation yields more intricate solutions [10]. Such situations arise naturally in the context of spectral matrices associated with reductions, for example, in the case of orthogonal Lie algebras.

We can now establish the following two propositions, as has been done in many related contexts (see, e.g. [8,10]).

Employing the formulas for the Darboux matrices S^+ and S^- in (36), one can derive their partial fraction decompositions, as summarized below.

Proposition 3.1: *Let S^+ and S^- be the Darboux matrices defined by (36). Then their partial fraction decompositions are given by*

$$S^+ = I_{2m} - \sum_{i=1}^N \frac{v_i^M \hat{v}_i}{\lambda - \hat{\lambda}_i}, \quad S^- = I_{2m} + \sum_{i=1}^N \frac{v_i \hat{v}_i^M}{\lambda - \lambda_i} \quad (43)$$

where the vectors v_i^M and $\hat{v}_i^M$ are defined via the $N \times N$ matrix M by

$$\begin{cases} (v_1^M, v_2^M, \dots, v_N^M) M = (v_1, v_2, \dots, v_N), \\ M((\hat{v}_1^M)^T, (\hat{v}_2^M)^T, \dots, (\hat{v}_N^M)^T)^T = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T. \end{cases} \quad (44)$$

From the above proposition, we deduce the following result.

Proposition 3.2: *For the Darboux matrices S^+ and S^- defined in (36), the following properties hold:*

$$S^+(\lambda_i)v_i = 0, \quad \hat{v}_i S^-(\hat{\lambda}_i) = 0, \quad 1 \leq i \leq N, \quad (45)$$

provided that $S^+(\lambda_i)$ and $S^-(\hat{\lambda}_i)$ are well-defined, and

$$S^+(\lambda)S^-(\lambda) = I_{2m}, \quad S_1^+ = -S_1^-, \quad (46)$$

provided the orthogonality condition

$$\hat{v}_i v_j = 0, \quad \text{if } \lambda_j = \hat{\lambda}_i, \quad \text{where } 1 \leq i, j \leq N, \quad (47)$$

is satisfied.

The condition (47) is required when eigenvalues coincide with adjoint eigenvalues. In particular, the first identity in (46) guarantees that $S^+(\lambda)$ and $S^-(\lambda)$ are mutual inverses.

For brevity, we introduce the block vectors:

$$v = (v_1, v_2, \dots, v_N), \quad \hat{v} = (\hat{v}_1^T, \hat{v}_2^T, \dots, \hat{v}_N^T)^T, \quad (48)$$

and the diagonal matrices

$$F = \text{diag} \left(\frac{1}{\lambda - \lambda_1}, \frac{1}{\lambda - \lambda_2}, \dots, \frac{1}{\lambda - \lambda_N} \right), \quad \hat{F} = \text{diag} \left(\frac{1}{\lambda - \hat{\lambda}_1}, \frac{1}{\lambda - \hat{\lambda}_2}, \dots, \frac{1}{\lambda - \hat{\lambda}_N} \right). \quad (49)$$

Then the Darboux matrices admit the compact form:

$$\begin{cases} S^+ = I_{2m} - vM^{-1}\hat{F}\hat{v}, \\ S^- = I_{2m} + vFM^{-1}\hat{v}. \end{cases} \quad (50)$$

One can verify the identity

$$\hat{v}v = \hat{F}^{-1}M - MF^{-1}, \quad (51)$$

which holds entrywise. Using this, a direct computation yields

$$S^+S^- = I_{2m} - vM^{-1}\hat{F}\hat{v} + vFM^{-1}\hat{v} - vM^{-1}\hat{F}\hat{v}vFM^{-1}\hat{v} = I_{2m}.$$

Taking the limits at infinity, we obtain

$$\begin{cases} S_1^+ = - \sum_{i,j=1}^N (M^{-1})_{ij} v_i \hat{v}_j = -vM^{-1}\hat{v}, \\ S_1^- = \sum_{i,j=1}^N (M^{-1})_{ij} v_i \hat{v}_j = vM^{-1}\hat{v}, \end{cases} \quad (52)$$

which immediately implies $S_1^+ = -S_1^-$.

We are ready to present a class of general binary DTs for the matrix integrable NLS equation (12), without imposing the skew-symmetric reductions on p and q .

Theorem 3.1: Let v_i and $\hat{v}_i$ ($1 \leq i \leq N$) be defined by (30) and (31), respectively, and let the M -matrix be given by (33), with the entries m_{ij}^c defined via (34), (35), and (32). If the

orthogonality condition (47) holds, then the matrix integrable NLS equation (12) admits the following binary DT:

$$\phi' = S^+ \phi, \quad \psi' = \psi S^-, \quad P' = P + S_1^+ \Lambda + \Lambda S_1^-, \quad (53)$$

where $S^\pm$ and $S_1^\pm$ are defined by (36) and (37), or equivalently by (50) and (52).

Proof: Assume that U' and V' are defined as in (40), with P' given by (41). The proof then reduces to verifying the two key conditions (38) and (39).

We first verify the spatial condition (38). From the spectral problems, we have

$$v_x = \Lambda v D + P v, \quad \hat{v}_x = -(\hat{D} \hat{v} \Lambda + \hat{v} P), \quad (54)$$

where

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N), \quad \hat{D} = \text{diag}(\hat{\lambda}_1, \hat{\lambda}_2, \dots, \hat{\lambda}_N). \quad (55)$$

Applying the definition of M in (33) together with (34) and (35), we obtain

$$M_x = \hat{v} \Theta v, \quad \Theta = (\Theta_{ij})_{N \times N} = (\Lambda)_{N \times N}. \quad (56)$$

Using the compact forms (50) and (52) for S^+ and $S_1^\pm$, together with the derivative formulas above, a straightforward calculation gives the spatial condition (38), where we have used the identities

$$\hat{D} = -\hat{F}^{-1} + \lambda I_N, \quad D = -F^{-1} + \lambda I_N, \quad (57)$$

and (51). Thus, the spatial condition is satisfied.

Next, for the temporal condition (39), we similarly have

$$v_t = \Lambda v D^2 + P v D + W^{[2]} v, \quad \hat{v}_t = -(\hat{D}^2 \hat{v} \Lambda + \hat{D} \hat{v} P + \hat{v} W^{[2]}), \quad (58)$$

with

$$W^{[2]} = V - \lambda^2 \Lambda - \lambda P = -\frac{1}{2} \Lambda (P^2 - P_x), \quad (59)$$

and

$$M_t = \hat{v} \Xi v, \quad \Xi = (\Xi_{ij})_{N \times N} = ((\lambda_j + \hat{\lambda}_i) \Lambda + P)_{N \times N} = \hat{D} \Lambda + \Lambda D + (P)_{N \times N}. \quad (60)$$

A direct, though lengthier, computation using the compact forms of S^+ and $S_1^\pm$, together with

$$P' = P + S_1^+ \Lambda - \Lambda S_1^+ = P - v M^{-1} \hat{v} \Lambda + \Lambda v M^{-1} \hat{v},$$

and the identities (57) and (51), shows that the temporal condition (39) also holds.

Hence, both spatial and temporal condition are satisfied, which completes the verification of the binary Darboux transformation. ■

3.2. The reduced case associated with $so(2m)$

The transformed potential P' defined by (41) must satisfy the orthogonal symmetry condition associated with $so(2m)$:

$$\Sigma P' \Sigma^{-1} = -(P')^T, \quad (61)$$

where Σ is defined in (18). Equivalently, in terms of the binary DT in (53) and using (46), this reduction condition can be expressed as

$$\Sigma [S_1^+, \Lambda] \Sigma^{-1} = [(S_1^+)^T, \Lambda], \quad (62)$$

where S_1^+ is given explicitly by (52).

To this end, we choose the adjoint eigenvalues

$$\hat{\lambda}_i = \lambda_i, \quad 1 \leq i \leq N, \quad (63)$$

and define the corresponding adjoint eigenfunctions, according to the symmetry of U and V , by

$$\hat{v}_i = v_i^T \Sigma^T = v_i^T \Sigma, \quad 1 \leq i \leq N, \quad (64)$$

where we have used the fact that $\Sigma^T = \Sigma$.

Furthermore, we impose the orthogonality condition

$$\hat{v}_j v_i = 0, \quad \text{whenever } \hat{\lambda}_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (65)$$

to ensure that the two Darboux matrices S^+ and S^- are inverses of each other.

Now, for $\hat{\lambda}_j \neq \lambda_i$, one can verify that

$$m_{ij} = \frac{\hat{v}_i v_j}{\lambda_j - \hat{\lambda}_i} = \frac{v_i^T \Sigma v_j}{\lambda_j - \hat{\lambda}_i} = -\frac{v_i^T \Sigma v_j}{\lambda_i - \hat{\lambda}_j} = -\frac{\hat{v}_j v_i}{\lambda_i - \hat{\lambda}_j} = -m_{ji}.$$

For $\hat{\lambda}_j = \lambda_i$, $1 \leq i, j \leq N$, we have

$$(m_{ij}^c + m_{ji}^c)_x = (m_{ij}^c + m_{ji}^c)_t = 0, \quad (66)$$

so that we can have $M^T = -M$. Then, from (52), it follows that

$$\Sigma S_1^+ \Sigma^{-1} = -(S_1^+)^T, \quad (67)$$

and together with

$$\Sigma \Lambda \Sigma^{-1} = -\Lambda, \quad (68)$$

this guarantees (62) and hence the reduction property (61) for the transformed potential matrix P' .

The result can be summarized as the following theorem.

Theorem 3.2: *Let the adjoint eigenvalues $\{\hat{\lambda}_i \mid 1 \leq i \leq N\}$ and the adjoint eigenfunctions $\{\hat{v}_i \mid 1 \leq i \leq N\}$ be defined by (63) and (64), respectively. Assume that M is defined by (33),*

and that whenever $\hat{\lambda}_j = \lambda_i$ for $1 \leq i, j \leq N$, the corresponding entries m_{ij}^c of M are determined by (34) and (35), satisfying

$$m_{ij}^c = -m_{ji}^c, \quad \text{when } \hat{\lambda}_j = \lambda_i, \quad 1 \leq i, j \leq N, \quad (69)$$

so that $M^T = -M$. If, in addition, the orthogonality conditions (65) hold, then the transformation (53) defines a binary DT for the skew-symmetric matrix integrable NLS model (12).

3.3. An illustrative example

We note that N must be even in order for M to be skew-symmetric and invertible. We now illustrate the construction with an example for $m=3$ and $N=2$, starting from the zero seed potential. At this moment, the eigenfunctions are

$$v_i = (e^{\lambda_i x + \lambda_i^2 t} \mu_1^{[i]T}, e^{-\lambda_i x - \lambda_i^2 t} \mu_2^{[i]T})^T, \quad 1 \leq i \leq 2, \quad (70)$$

and the adjoint eigenfunctions are

$$\hat{v}_i = v_i^T \Sigma, \quad 1 \leq i \leq 2. \quad (71)$$

Here $\mu_j^{[i]} \in \mathbb{C}^3$, $1 \leq i, j \leq 2$, are constant column vectors.

Since $\hat{\lambda}_i = \lambda_i$, $1 \leq i \leq 2$, the orthogonality conditions

$$\hat{v}_i v_i = v_i^T \Sigma v_i = 0, \quad 1 \leq i \leq 2, \quad (72)$$

must be satisfied. These are equivalent to

$$\mu_1^{[i]T} \mu_2^{[i]} = 0, \quad 1 \leq i \leq 2. \quad (73)$$

If we take $m_{11}^c = m_{22}^c = 0$, two more sets of orthogonality conditions must also be considered. The first set,

$$\hat{v}_i \Lambda v_i = 0, \quad 1 \leq i \leq 2, \quad (74)$$

is automatically satisfied. The second set,

$$2\lambda_i \hat{v}_i \Lambda v_i = 0, \quad 1 \leq i \leq 2, \quad (75)$$

which follows directly from (74). Consequently, the only the conditions in (73) need to be imposed.

It is then straightforward to compute from

$$p' = [S_1^+, \Lambda]_{12} = -2(S_1^+)_{12},$$

that

$$\begin{cases} p'_{12} = -p'_{21} = \frac{2(\lambda_1 - \lambda_2)(\mu_{1,1,1}\mu_{2,1,2} - \mu_{1,1,2}\mu_{2,1,1})}{\mu_2^{[1]T}\mu_1^{[2]}e^{-2(\lambda_1 x + \lambda_1^2 t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{-2(\lambda_2 x + \lambda_2^2 t)}}, \\ p'_{13} = -p'_{31} = \frac{2(\lambda_1 - \lambda_2)(\mu_{1,1,1}\mu_{2,1,3} - \mu_{1,1,3}\mu_{2,1,1})}{\mu_2^{[1]T}\mu_1^{[2]}e^{-2(\lambda_1 x + \lambda_1^2 t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{-2(\lambda_2 x + \lambda_2^2 t)}}, \\ p'_{23} = -p'_{32} = \frac{2(\lambda_1 - \lambda_2)(\mu_{1,1,2}\mu_{2,1,3} - \mu_{1,1,3}\mu_{2,1,2})}{\mu_2^{[1]T}\mu_1^{[2]}e^{-2(\lambda_1 x + \lambda_1^2 t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{-2(\lambda_2 x + \lambda_2^2 t)}}. \end{cases} \quad (76)$$

Similarly, from

$$q' = [S_1^+, \Lambda]_{21} = 2(S_1^+)_{21},$$

we have

$$\begin{cases} q'_{12} = -q'_{21} = -\frac{2(\lambda_1 - \lambda_2)(\mu_{1,2,1}\mu_{2,2,2} - \mu_{1,2,2}\mu_{2,2,1})}{\mu_2^{[1]T}\mu_1^{[2]}e^{2(\lambda_2x+\lambda_2^2t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{2(\lambda_1x+\lambda_1^2t)}}, \\ q'_{13} = -q'_{31} = -\frac{2(\lambda_1 - \lambda_2)(\mu_{1,2,1}\mu_{2,2,3} - \mu_{1,2,3}\mu_{2,2,1})}{\mu_2^{[1]T}\mu_1^{[2]}e^{2(\lambda_2x+\lambda_2^2t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{2(\lambda_1x+\lambda_1^2t)}}, \\ q'_{23} = -q'_{32} = -\frac{2(\lambda_1 - \lambda_2)(\mu_{1,2,2}\mu_{2,2,3} - \mu_{1,2,3}\mu_{2,2,2})}{\mu_2^{[1]T}\mu_1^{[2]}e^{2(\lambda_2x+\lambda_2^2t)} + \mu_1^{[1]T}\mu_2^{[2]}e^{2(\lambda_1x+\lambda_1^2t)}}. \end{cases} \quad (77)$$

Here, we assume that S_1^+ is partitioned in the same manner as U , and that

$$p' = (p'_{ij})_{3 \times 3}, \quad q' = (q'_{ij})_{3 \times 3}, \quad \mu_s^{[r]} = (\mu_{r,s,1}, \mu_{r,s,2}, \mu_{r,s,3})^T, \quad 1 \leq r, s \leq 2. \quad (78)$$

Under the conditions in (73), these functions provide explicit solutions to the matrix integrable NLS model with skew-symmetric matrix potentials in (12).

4. Concluding remarks

This paper investigates binary DTs for the integrable NLS equation with two skew-symmetric matrix potentials, using the extended M -matrix generating scheme. The resulting DT structures play a fundamental role for exploring the integrable dynamics of the considered matrix NLS model. Starting from the vacuum state, repeated applications of these binary DTs yield a wide variety of explicit, exact soliton solutions. The concrete example presented here illustrate the diversity of coupled nonlinear integrable systems, demonstrating the practical applicability of the framework and the rich integrability structures inherent in such systems. In particular, the M -matrix must be extended to handle cases where eigenvalues coincide with adjoint eigenvalues, for which the classical M -matrix formulation becomes inapplicable.

By examining the eigenvalues of the associated spectral problems, soliton solutions can be classified into types such as negatons, positons, and complexitons [26,27], and further categorized into solitons, breathers, kinks, anti-kinks, lump waves, rogue waves, and various mixed interaction solutions. Interaction solutions may even involve arbitrary functions (see, e.g. [28]). Particular solutions can be obtained via reductions of wave numbers from general soliton solutions (see, e.g. [29,30]). Lump wave solutions, in particular, have been extensively studied in both linear and nonlinear models (see, e.g. [31,32]).

We further note that increasing the number of dependent variables or including higher powers of the spectral parameter in the spatial spectral matrix can generate larger and more complex integrable systems (see, e.g. [33,34]), including integrable couplings (see, e.g. [35,36]). Such complexity, especially with the emergence of non-localities, introduces additional analytical challenges. Nevertheless, classifying these extended integrable systems and studying their dynamical properties will enhance understanding of the fundamental mechanisms underlying nonlinear dynamics and integrability in mathematical physics [37,38].

Author contributions

CRedit: **Wen-Xiu Ma**: Conceptualization, Investigation, Methodology, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing

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References

- [1] Lax PD. Integrals of nonlinear equations of evolution and solitary waves. *Comm Pure Appl Math.* 1968;21(5):467–490. doi: [10.1002/cpa.3160210503](https://doi.org/10.1002/cpa.3160210503)
- [2] Ablowitz MJ, Segur H. Solitons and the inverse scattering transform. Philadelphia (PA): SIAM; 1981.
- [3] Gu CH, Hu HS, Zhou ZX. Darboux transformations in integrable systems. Dordrecht: Springer; 2005.
- [4] Novikov SP, Manakov SV, Pitaevskii LP, et al. Theory of solitons: the inverse scattering method. New York: Consultants Bureau; 1984.
- [5] Tu GZ. On Liouville integrability of zero-curvature equations and the Yang hierarchy. *J Phys A: Math Gen.* 1989;22(13):2375–2392. doi: [10.1088/0305-4470/22/13/031](https://doi.org/10.1088/0305-4470/22/13/031)
- [6] Ma WX, Chen M. Hamiltonian and quasi-Hamiltonian structures associated with semi-direct sums of Lie algebras. *J Phys A: Math Gen.* 2006;39(34):10787–10801. doi: [10.1088/0305-4470/39/34/013](https://doi.org/10.1088/0305-4470/39/34/013)
- [7] Ma WX, Batwa S. A binary Darboux transformation for multicomponent NLS equations and their reductions. *Anal Math Phys.* 2021;11(2):44. doi: [10.1007/s13324-021-00477-5](https://doi.org/10.1007/s13324-021-00477-5)
- [8] Ma WX. A novel kind of reduced integrable matrix mKdV equations and their binary Darboux transformations. *Mod Phys Lett B.* 2022;36(20):2250094. doi: [10.1142/S0217984922500944](https://doi.org/10.1142/S0217984922500944)
- [9] Müller-Hoissen F. A vectorial binary Darboux transformation for the first member of the negative part of the AKNS hierarchy. *J Phys A: Math Theoret.* 2023;56(12):125701. doi: [10.1088/1751-8121/ac980d](https://doi.org/10.1088/1751-8121/ac980d)
- [10] Ma WX. Binary Darboux transformation of vector nonlocal reverse-time integrable NLS equations. *Chaos Solitons Fractals.* 2024;180:114539. doi: [10.1016/j.chaos.2024.114539](https://doi.org/10.1016/j.chaos.2024.114539)
- [11] Ma WX. An application of dual group reductions to the AKNS integrable mKdV model. *Mod Phys Lett B.* 2025;39(35):2550233. doi: [10.1142/S0217984925502331](https://doi.org/10.1142/S0217984925502331)
- [12] Ma WX. The algebraic structure of zero curvature representations and application to coupled KdV systems. *J Phys A: Math Gen.* 1993;26(11):2573–2582. doi: [10.1088/0305-4470/26/11/009](https://doi.org/10.1088/0305-4470/26/11/009)
- [13] Guo FK. A variant of Lax representations and Lax representations of hierarchies of Hamilton's equations. *Acta Math Sinica.* 1994;37(4):515–522. doi: [10.12386/A1994sxxb0069](https://doi.org/10.12386/A1994sxxb0069)
- [14] Fordy AP, Kulish PP. Nonlinear Schrödinger equations and simple Lie algebras. *Commun Math Phys.* 1983;89(3):427–443. doi: [10.1007/BF01214664](https://doi.org/10.1007/BF01214664)
- [15] Geng XG, Li RM, Xue B. A vector general nonlinear Schrödinger equation with $(m+n)$ components. *J Nonlinear Sci.* 2020;30(3):991–1013. doi: [10.1007/s00332-019-09599-4](https://doi.org/10.1007/s00332-019-09599-4)
- [16] Ma WX. Integrable nonlocal nonlinear Schrödinger equations associated with $so(3, R)$. *Proc Amer Math Soc Ser B.* 2022;9:1–11. doi: [10.1090/bproc/116](https://doi.org/10.1090/bproc/116)
- [17] Wang HF, Zhang YF. Application of Riemann–Hilbert method to an extended coupled nonlinear Schrödinger equations. *J Comput Appl Math.* 2023;420:114812. doi: [10.1016/j.cam.2022.114812](https://doi.org/10.1016/j.cam.2022.114812)

- [18] Zhou RG, Zhu HY. An integrable matrix NLS equation on star graph and symmetry-dependent connection conditions of vertex. *Comput Appl Math.* **2023**;42(2):69. doi: [10.1007/s40314-023-02201-4](https://doi.org/10.1007/s40314-023-02201-4)
- [19] Ma WX, Khalique CM. Integrable reductions of matrix AKNS soliton hierarchies. *Rep Math Phys.* **2026**;97(3):275–286. doi: [10.1016/S0034-4877\(26\)00035-2](https://doi.org/10.1016/S0034-4877(26)00035-2)
- [20] Hasegawa A. Optical solitons in fibers. Berlin: Springer-Verlag; **1990**.
- [21] Matveev VB, Salle MA. Darboux transformations and solitons. Berlin: Springer-Verlag; **1991**.
- [22] Ma WX. Darboux transformations for a Lax integrable system in $2n$ dimensions. *Lett Math Phys.* **1997**;39(1):33–49. doi: [10.1007/s11005-997-3049-3](https://doi.org/10.1007/s11005-997-3049-3)
- [23] Ma WX, Zhang YJ. Darboux transformations of integrable couplings and applications. *Rev Math Phys.* **2018**;30(2):1850003. doi: [10.1142/S0129055X18500034](https://doi.org/10.1142/S0129055X18500034)
- [24] Terng CL, Wu ZW. Darboux transformations for the $\hat{A}_{2n}^{(2)}$ -KdV hierarchy. *J Geom Anal.* **2023**;33:111. doi: [10.1007/s12220-022-01158-w](https://doi.org/10.1007/s12220-022-01158-w)
- [25] Yang J. Nonlinear waves in integrable and nonintegrable systems. Philadelphia: SIAM; **2010**.
- [26] Ma WX, You Y. Solving the Korteweg-de Vries equation by its bilinear form: Wronskian solutions. *Trans Amer Math Soc.* **2005**;357(5):1753–1778. doi: [10.1090/S0002-9947-04-03726-2](https://doi.org/10.1090/S0002-9947-04-03726-2)
- [27] Ma WX. Complexiton solutions to integrable equations. *Nonlinear Anal.* **2005**;63(5-7):e2461–e2471. doi: [10.1016/j.na.2005.01.068](https://doi.org/10.1016/j.na.2005.01.068)
- [28] Ma WX, Yong XL, Zhang HQ. Diversity of interaction solutions to the $(2 + 1)$ -dimensional ito equation. *Comput Math Appl.* **2018**;75(1):289–295. doi: [10.1016/j.camwa.2017.09.013](https://doi.org/10.1016/j.camwa.2017.09.013)
- [29] Guo BL, Ling LM, Liu QP. High-order solutions and generalized Darboux transformations of derivative nonlinear Schrödinger equations. *Stud Appl Math.* **2013**;130(4):317–344. doi: [10.1111/j.1467-9590.2012.00568.x](https://doi.org/10.1111/j.1467-9590.2012.00568.x)
- [30] Yang SX, Wang YF, Zhang X. Conservation laws, Darboux transformation and localized waves for the N -coupled nonautonomous gross-Pitaevskii equations in the Bose-Einstein condensates. *Chaos Solitons Fractals.* **2023**;169:113272. doi: [10.1016/j.chaos.2023.113272](https://doi.org/10.1016/j.chaos.2023.113272)
- [31] Ma WX. Dispersion-governed lump waves in a generalized Calogero Bogoyavlenskii Schiff-like model with spatially symmetric nonlinearity. *Axioms.* **2025**;14(12):869. doi: [10.3390/axioms14120869](https://doi.org/10.3390/axioms14120869)
- [32] Ma WX. Lump waves and their dynamics in a generalized Kadomtsev–Petviashvili-like model. *Mod Phys Lett A.* **2026**;41(1):2550215. doi: [10.1142/S0217732325502153](https://doi.org/10.1142/S0217732325502153)
- [33] Zhang YF, Han Z, Zhao ZL. Applications of a few Lie algebras. *Acta Math Appl Sin Engl Ser.* **2016**;32(2):289–304. doi: [10.1007/s10255-016-0553-1](https://doi.org/10.1007/s10255-016-0553-1)
- [34] Geng XG, Jia MX, Xue B, et al. Application of tetragonal curves to coupled Boussinesq equations. *Lett Math Phys.* **2024**;114(1):30. doi: [10.1007/s11005-024-01780-5](https://doi.org/10.1007/s11005-024-01780-5)
- [35] Ma WX, Xu XX, Zhang YF. Semi-direct sums of Lie algebras and continuous integrable couplings. *Phys Lett A.* **2006**;351(3):125–130. doi: [10.1016/j.physleta.2005.09.087](https://doi.org/10.1016/j.physleta.2005.09.087)
- [36] Xu XX. An integrable coupling hierarchy of the MkdV integrable systems, its Hamiltonian structure and corresponding nonisospectral integrable hierarchy. *Appl Math Comput.* **2010**;216(1):344–353. doi: [10.1016/j.amc.2010.01.016](https://doi.org/10.1016/j.amc.2010.01.016)
- [37] Ma WX. Nonlocality, integrability, and solitons. *Theoret Math Phys.* **2025**;224(1):1220–1233. doi: [10.1134/S0040577925070104](https://doi.org/10.1134/S0040577925070104)
- [38] Ma WX, Khalique CM. Matrix integrable models associated with reduced AKNS Lax pairs. *J Geom Phys.* **2026**;222:105772. doi: [10.1016/j.geomphys.2026.105772](https://doi.org/10.1016/j.geomphys.2026.105772)